RESULTS SUMMARY — FOR THESIS INCLUSION

Leaf Segmentation and Growth-Stage Classification: U-Net Performance and Fusion Pipeline Evaluation

Branch 01 (U-Net) standalone results, and its performance as the segmentation stage of the U-Net + ResNet50 fusion pipeline, with cross-validation.

Contents

1. U-Net Segmentation Model

1.1 Architecture and Training Configuration

A U-Net (Ronneberger et al., 2015) encoder-decoder network was implemented and trained from scratch, without a pretrained backbone, for binary segmentation of tomato leaves from background.

Table 1. U-Net architecture and training configuration.

Encoder	4 stages of two 3×3 convolutions (BatchNorm, ReLU) + max-pool; 32→64→128→256 channels
Bottleneck	512 channels
Decoder	4 stages of transposed-convolution upsampling, skip connections, two 3×3 convolutions; 256→128→64→32 channels
Output	1×1 convolution, 1-channel logits; sigmoid + 0.5 threshold applied at inference
Parameters	7.77 million
Input / output resolution	224 × 224 × 3 → 224 × 224 × 1
Loss function	0.5 × Binary Cross-Entropy + 0.5 × (1 − Dice coefficient)
Optimizer	Adam, learning rate 1×10^{-4}, `ReduceLROnPlateau` (factor 0.5, patience 5, monitored on validation Dice)
Batch size	8
Epochs	60 (best checkpoint selected by validation Dice)
Data augmentation	Horizontal flip, vertical flip, 90°-multiple rotation — applied identically to image and mask
Dataset split	687 train / 135 validation / 160 test images (982 total, four growth stages)

1.2 Overlap-Based Performance (Dice, IoU)

The model was trained twice under identical configuration to verify reproducibility. Both runs converged to statistically indistinguishable results.

Table 2. Segmentation Dice and IoU, both training runs.

Run	Best epoch	Val. Dice	Val. IoU	Test Dice	Test IoU
Run 1	59 / 60	0.8134	0.7289	0.7534	0.6642
Run 2 (reported)	**58 / 60**	**0.8127**	**0.7241**	**0.7572**	**0.6650**

Table 3. Test-set Dice and IoU by growth stage (160 test images).

Growth stage	Dice	IoU	n
Seeding	**0.858**	**0.776**	**38**

Growth stage	Dice	IoU	n
Developing	0.774	0.687	41
Flowering	0.729	0.631	60
Fruiting	0.625	0.519	21

> Corresponding figure: `outputs/training_curves.png` (loss/Dice/IoU per epoch, train vs. validation); qualitative image/ground-truth/prediction examples: `outputs/qualitative_examples/test_examples.png`.

1.3 Pixel-Level Classification Performance

Dice and IoU describe mask overlap but are not, by themselves, standard classification metrics. To report accuracy, precision, recall, and F1-score, every pixel across all 160 test images was treated as an independent binary classification instance (leaf vs. background), and scored against the model's final exported checkpoint.

Table 4. Pixel-level classification metrics, aggregated over all test-set pixels (8,028,160 pixels total).

Class	Precision	Recall	F1-score	Support (pixels)
Background	0.9701	0.9506	0.9603	6,835,618
Leaf	0.7462	0.8320	0.7868	1,192,542

Overall pixel accuracy: 0.9330. Leaf pixels make up approximately 14.9% of the test set, so accuracy alone is inflated by the majority background class; precision, recall, and F1 for the leaf class are the more informative figures.

Table 5. Confusion matrix, pixel level (rows = ground truth, columns = predicted).

	Predicted background	Predicted leaf
Actual background	6,498,216	337,402
Actual leaf	200,310	992,232

Table 6. Leaf-class precision, recall, and F1-score by growth stage (mean over images in that stage).

Growth stage	Precision	Recall	F1-score
Seeding	**0.8384**	**0.9129**	**0.8702**
Developing	0.7744	0.8561	0.7994
Flowering	0.7224	0.8313	0.7661
Fruiting	0.6600	0.6938	0.6544

FINDING

Fruiting is consistently the weakest growth stage across both overlap-based (Table 3) and classification-based (Table 6) metrics. It is the smallest class in the dataset (21 of 160 test images) and leaves are frequently occluded by fruit clusters — a property of the data, not a preprocessing or architecture limitation.

1.4 Cross-Validation

To assess sensitivity to the specific train/validation split, 5-fold stratified cross-validation (by growth stage) was performed over the pooled train+validation set (822 images), with the original held-out test set (160 images) never included in any fold. Each fold retrained U-Net from scratch. An initial run at 8 epochs/fold was found to be substantially undertrained relative to the 60-epoch single-split model and was discarded in favor of a 30-epoch/fold run, reported below.

Table 7. Per-fold segmentation results, 30 epochs/fold, evaluated on the fixed held-out test set.

Fold	Best val. Dice	Test accuracy	Leaf precision	Leaf recall	Leaf F1
1	0.7337	0.9307	0.7379	0.8274	0.7801
2	0.7237	0.9268	0.7273	0.8120	0.7673
3	0.7092	0.9199	0.6787	0.8753	0.7646
4	0.6847	0.9277	0.7277	0.8198	0.7710
5	0.7380	0.9347	0.7724	0.7945	0.7833
Mean ± SD	**0.7179 ± 0.0216**	**0.9280 ± 0.0054**	**0.7288 ± 0.0335**	**0.8258 ± 0.0302**	**0.7732 ± 0.0081**

FINDING

Cross-validation F1 (0.7732 ± 0.0081) is closely comparable to, and marginally exceeds, the single-split test F1 (0.7868 leaf-F1 in Table 4, computed at pixel level; 0.7572 Dice in Table 2), with a small standard deviation across folds — indicating the segmentation model's performance is stable and not an artifact of one favorable data split. Best validation Dice (0.7179 ± 0.0216) remained below the fully-converged single-split model's 0.8127, indicating folds trained for 30 epochs had not fully converged relative to the 60-epoch benchmark; this affects Section 2.4 more than the figures reported here.

2. Fusion Pipeline (U-Net + ResNet50)

2.1 Pipeline Methodology

The fusion pipeline segments a raw photograph with U-Net, uses the predicted binary mask to zero out the background, and classifies the masked image into one of four growth stages with a ResNet50 (ImageNet-pretrained, fully fine-tuned) trained specifically on background-masked images. The two models are trained entirely independently and combined only at inference time.

An important methodological finding emerged during pipeline evaluation. U-Net was originally trained and evaluated using a direct resize to 224×224 (distorting aspect ratio). The classifier, however, expects `Resize(256) → CenterCrop(224)` (aspect-preserving). Feeding U-Net its native preprocessing frame at pipeline inference time produced a predicted mask that was not geometrically aligned with what the classifier actually received, despite the mask itself being accurate in U-Net's own frame. Re-aligning U-Net's inference preprocessing to match the classifier's frame — with no change to either model's weights — corrected this mismatch. All results in Sections 2.2–2.4 use this crop-aligned configuration.

2.2 Single-Split Test Performance

Table 8. Full-pipeline classification report, held-out test set (160 images).

Growth stage	Precision	Recall	F1-score	Support
Developing	0.9512	0.9512	0.9512	41
Flowering	0.8644	0.8500	0.8571	60

Growth stage	Precision	Recall	F1-score	Support
Fruiting	0.6818	0.7143	0.6977	21
Seeding	**1.0000**	**1.0000**	**1.0000**	**38**
Macro average	*0.8744*	*0.8789*	*0.8765*	160
Weighted average	*0.8949*	*0.8938*	*0.8942*	160

Overall accuracy: 0.8938 (89.38%).

Table 9. Full-pipeline confusion matrix (rows = true stage, columns = predicted stage).

	Developing	Flowering	Fruiting	Seeding
Developing	39	2	0	0
Flowering	2	51	7	0
Fruiting	0	6	15	0
Seeding	0	0	0	38

The dominant error mode is confusion between flowering and fruiting (7 flowering images predicted as fruiting; 6 fruiting images predicted as flowering) — the two visually and temporally adjacent growth stages, and the two stages most affected by leaf occlusion (by flowers and fruit clusters, respectively).

> Corresponding figure: `outputs/fusion_pipeline/full_pipeline_confusion_matrix.png`; visual walkthrough (input / predicted mask / masked image / class probabilities, one example per stage): `outputs/fusion_pipeline/pipeline_walkthrough.png`.

2.3 Segmenter Comparison

The U-Net segmenter was compared against a DeepLabV3-ResNet50 segmenter trained on the same data and fed into the identical classifier, isolating the effect of the segmentation model choice (and the preprocessing fix described in Section 2.1) on full-pipeline accuracy.

Table 10. Segmenter comparison — segmentation quality vs. full-pipeline accuracy, same test set and classifier throughout.

Segmenter	Seg. foreground IoU	Seg. foreground Dice	Full-pipeline accuracy
DeepLabV3-ResNet50	0.623	0.757	0.8438
U-Net (native preprocessing)	0.665	0.784	0.7875
U-Net (crop-aligned preprocessing)	**0.667**	**0.783**	**0.8938**

FINDING

U-Net's own segmentation quality (IoU/Dice) is superior to DeepLabV3's under both preprocessing conventions, yet under its native preprocessing the full pipeline scored lower than DeepLabV3's pipeline (0.7875 vs. 0.8438). Correcting only the preprocessing alignment — no retraining, no architecture change — raised U-Net's pipeline accuracy to 0.8938, the best of all three configurations, while segmentation IoU/Dice remained essentially unchanged (0.665→0.667 IoU). This demonstrates that the initial deficit was attributable entirely to a geometric mismatch between the segmentation and classification stages, not to segmentation quality.

2.4 Cross-Validation

For each of the five U-Net folds trained in Section 1.4, the full pipeline was evaluated against the fixed test set using the same fixed classifier throughout (the classifier was not retrained per fold). This isolates the segmenter's contribution to pipeline accuracy variance.

Table 11. Per-fold full-pipeline results, 30 epochs/fold.

Fold	Accuracy	Macro precision	Macro recall	Macro F1
1	0.8500	0.8334	0.8536	0.8386
2	0.8625	0.8545	0.8561	0.8525
3	0.8313	0.8161	0.8527	0.8228
4	**0.8938**	**0.8739**	**0.8944**	**0.8809**
5	0.8438	0.8220	0.8378	0.8254
Mean ± SD	**0.8562 ± 0.0238**	**0.8400 ± 0.0239**	**0.8589 ± 0.0211**	**0.8441 ± 0.0237**

Table 12. Full-pipeline per-class F1-score across folds, mean ± SD.

Growth stage	Precision	Recall	F1-score
Developing	0.9474 ± 0.0432	0.9171 ± 0.0223	0.9305 ± 0.0315
Flowering	0.8455 ± 0.0298	0.7567 ± 0.1147	0.7966 ± 0.0415
Fruiting	0.5671 ± 0.0819	0.7619 ± 0.0521	0.6491 ± 0.0586
Seeding	**1.0000 ± 0.0000**	**1.0000 ± 0.0000**	**1.0000 ± 0.0000**

FINDING

Cross-validated full-pipeline accuracy (0.8562 ± 0.0238) is lower than the single-split result reported in Section 2.2 (0.8938) by 3.76 percentage points. An earlier cross-validation run at 8 epochs/fold produced a substantially wider gap (0.8075 ± 0.0436, 8.63 points below the single-split figure); increasing training to 30 epochs/fold recovered most, but not all, of this gap. Because best validation Dice at 30 epochs/fold (0.7179, Table 7) remained below the fully-converged single-split model's 0.8127, some residual undertraining cannot be ruled out as a contributor to the remaining gap. **The cross-validated mean of 85.62% (± 2.38 percentage points) is considered the more statistically defensible estimate of expected pipeline accuracy; the single-split figure of 89.38% represents a best-case result from one particular data partition** and should be reported alongside, not in place of, the cross-validated estimate.

Corresponding figure: `outputs/cross_validation/cv_summary.png` (per-fold segmentation F1 and pipeline accuracy, with mean and single-split baseline overlaid). Full per-fold data: `outputs/cross_validation/unet_segmentation_cv_results.csv`, `outputs/cross_validation/fusion_pipeline_cv_results.csv`.

3. Challenges Encountered

1. **DataLoader multiprocessing instability (Windows/Jupyter).** PyTorch `DataLoader` worker subprocesses (`num_workers > 0`) hung indefinitely under Windows in a Jupyter kernel, with no exception raised — training appeared to run but produced no output for hours. Resolved

by setting `num_workers=0` (in-process data loading), which was fast enough at this dataset scale (982 images) not to be a bottleneck.

2. **Preprocessing-geometry mismatch between segmentation and classification stages.** Described in detail in Section 2.1 and 2.3 — the single largest source of full-pipeline accuracy variance encountered in this research (10.6 percentage points), caused not by model quality but by a frame mismatch between the two independently-developed models.
3. **Class imbalance.** The fruiting stage is represented by only 21 of 160 test images (13.1%) and 71–101 of the corresponding train/pool images across splits — the smallest class throughout the dataset. Combined with visual occlusion of leaves by fruit clusters, this makes fruiting the weakest-performing class in every experiment reported here, at both the segmentation and classification stages.
4. **Reproducibility and checkpoint provenance.** Editor/kernel state desynchronization during iterative development (a Jupyter notebook's in-memory kernel state and its on-disk file can silently diverge) required deliberate reconciliation steps to ensure reported results matched the code and configuration actually described.
5. **Cross-validation compute budget.** Full 5-fold cross-validation at the single-split model's training length (60 epochs/fold) was judged too time-costly to run initially; an 8-epoch/fold pilot run was found to be substantially undertrained (Section 1.4, Section 2.4) and was superseded by a 30-epoch/fold run. The gap between the 30-epoch cross-validated result and the fully-converged single-split result suggests full parity would require the complete 60-epoch/fold budget.
6. **Background dependency of the classifier.** A separate ablation study (outside the scope of this document, reported alongside the ResNet50 branch) found that a classifier trained on masked images performs substantially worse on unmasked images (35.6% accuracy) than a classifier trained on unmasked images performs on masked images (76.9%) — establishing that the fusion pipeline must mask consistently at both training and inference time, which the pipeline described in Section 2 does.

4. Summary of Findings

- U-Net achieves **0.7572 test Dice / 0.6650 test IoU** and **0.7868 leaf-class F1 (0.7462 precision, 0.8320 recall)** at the pixel level on a held-out test set, with 5-fold cross-validated leaf F1 of **0.7732 ± 0.0081** confirming stability across data splits.

- The fusion pipeline (U-Net + ResNet50) achieves **89.38% single-split accuracy** (macro F1 0.8765), and a cross-validated mean accuracy of **85.62% ± 2.38 percentage points**, which is recommended as the more defensible figure for reporting expected real-world performance.
- A preprocessing-alignment fix between the segmentation and classification stages — not a change to either model — improved full-pipeline accuracy by 10.6 percentage points (78.75% → 89.38%), and is the single most consequential methodological finding of this evaluation.
- Fruiting is the weakest growth stage across every metric and every experiment (segmentation Dice/IoU, pixel classification F1, and full-pipeline classification F1), attributable to class scarcity and visual occlusion by fruit — a data limitation rather than a modeling one.

All figures computed from the current exported checkpoints (`export/unet_tomato_final.pth`, `checkpoints_masked/resnet50_masked_best.pt`) and the fixed 160-image held-out test set. Source notebooks: `unet_pipeline.ipynb`, `unet_resnet_fusion_pipeline.ipynb`, `unet_cross_validation.ipynb`.